Unit: 6 Unit 2: Exponents Radicals
Topic: Exponents

Name: ________________
Date: __________

Homework (Jalapeno)

Q1. The ancient Egyptians used 2^3 to represent the number of sides of a triangle minus 2. What is 2^3 equal to?
(A) 6
(B) 8
(C) 4
(D) 5
(E) 3

Q2. The ancient Greek mathematician Archimedes used the expression $3^2 \cdot 2^2$ to calculate the area of a square. What is $3^2 \cdot 2^2$ equal to?
(A) 36
(B) 24
(C) 20
(D) 18
(E) 12

Q3. The Rosetta Stone has an inscription with the expression x^2 where $x = 4$. What is the value of x^2?
(A) 12
(B) 16
(C) 14
(D) 10
(E) 8

Q4. The ancient Babylonians used the expression $(2^2)^3$ to represent the volume of a cube. What is $(2^2)^3$ equal to?
(A) 64
(B) 32
(C) 16
(D) 8
(E) 4

Q5. The ancient Chinese mathematician Liu Hui used the expression $4^2 \div 2^2$ to calculate the ratio of areas. What is $4^2 \div 2^2$ equal to?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 12

Q6. The Great Pyramid of Giza has a height that can be represented by the expression $3^2 + 2^2$. What is $3^2 + 2^2$ equal to?
(A) 13
(B) 12
(C) 11
(D) 10
(E) 9

Q7. The ancient Indian mathematician Aryabhata used the expression $2^4 \cdot 3^2$ to calculate the area of a rectangle. What is $2^4 \cdot 3^2$ equal to?
(A) 144
(B) 128
(C) 120
(D) 100
(E) 80

Q8. The ancient Greek mathematician Euclid used the expression $(3^2)^2$ to represent the area of a square. What is $(3^2)^2$ equal to?
(A) 81
(B) 64
(C) 36
(D) 27
(E) 24

Open Ended Questions (FRQ)

Q9. The ancient Egyptians built a pyramid with a base area of 4^2 square meters and a height of 2^3 meters. Calculate the volume of the pyramid in cubic meters, given that the formula for the volume of a pyramid is $\frac{1}{3} \cdot (\textit{base area}) \cdot \textit{height}$. Show all your work and calculations.

Q10. The ancient Babylonians used the expression x^3 to represent the volume of a cube, where x is the length of a side. If the volume of the cube is 64 cubic meters, find the value of x and show all your work and calculations.

Chef's Tips

- Emphasize the importance of using exponent rules to simplify expressions
- Encourage students to use historical and ancient contexts to understand the relevance of exponents
- Remind students to show all work and calculations for free response questions